Kaixiang Yao

Robotics Engineering Undergraduate · Southern University of Science and Technology

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EDUCATION

Southern University of Science and Technology Shenzhen, China · Aug. 2023-Jun. 2027

B.Eng. in Robotic Engineering · GPA: 3.91/4.0 · Major Rank: 1/90

Selected coursework: Fundamentals of Control Engineering (100), Mechanics of Materials (100), Probability & Mathematical Statistics (99), Theory of Machines (98), Robot Operating System (97), Mechanical Design (96), Robot Modeling and Control (94).

University of California, Berkeley Berkeley, CA · Jun.-Aug. 2026

Summer Session

RESEARCH FOCUS

> Physically grounded robot learning: using contact, deformation, force, and task outcomes to make physical interaction safer and more adaptive.

PUBLICATIONS & MANUSCRIPTS

01 **Kaixiang Yao**, Zeyang Wang, Jinxin Sun, Kong Hoi Cheng, Junyun Fu, Zhaokai Chen, Chengyu Lin, Chenglong Fu, "Gaze-Driven Perception-Action Control of Multi-Channel FES for Efficient Grasp Restoration," *under revision at IEEE Transactions on Medical Robotics and Bionics*, 2026.

02 **Kaixiang Yao**, Hao Chi, Dingye Chen, Jiwen Zhang, "Quasi-Real-Time Digital Twin for Robotic Assembly," *Journal of Intelligent Manufacturing*, 2026.

03 **Kaixiang Yao**, Chao Huang, Yanlin Liu, Yao Shen, "CARTILAGE: A Convergence-Aware Rigid-Soft Contact Runtime for Batched Robot Learning," *under review at Conference on Robot Learning (CoRL)*, 2026.

RESEARCH EXPERIENCE

University of California, Berkeley Berkeley, CA · Jun. 2026-Present

Visiting Undergraduate Researcher · Advised by Prof. Masayoshi Tomizuka · Mentored by Yuxin Chen

Embodied AI for Physical Manipulation

- Developing simulation-based evaluation workflows for learning and control in contact-rich manipulation.
- Studying how physical interaction feedback can support robust behavior under changing dynamics.
- Designing controlled evaluations to identify failure modes and improve experimental rigor.

Shanghai Jiao Tong University Shanghai, China · Oct. 2025-Present

Undergraduate Research Assistant · Advisor: Prof. Yao Shen

Robot Learning for High-Fidelity Rigid-Soft Contact

- Developed a simulation-and-learning framework for a four-finger PneuNet soft gripper and fragile-object grasping.
- Designed soft-gripper assemblies, pneumatic actuation semantics, contact/tactile definitions, and RL-compatible tasks.
- Studied how contact fidelity and force-safety rewards affect damage-free grasping.

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Research Experience, Skills & Recognition
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RESEARCH EXPERIENCE

Southern University of Science and Technology — Shenzhen, China · Jul. 2024-May 2026

Undergraduate Research Assistant · Advisor: Prof. Chenglong Fu

Eye-Tracking and Electrical Stimulation-Assisted Grasping

- Proposed normalized gaze-variance features and random-forest recognition with dual-threshold temporal filtering.
- Integrated YOLOv5 object detection and MediaPipe hand-keypoint estimation for object-aware grasping strategies.
- Designed an 8-channel, 54-electrode high-density FES sleeve for selective muscle activation.
- Verified approximately constant interaction time as gesture options increased, supporting O(1)-style selection behavior.

Tsinghua University — Beijing, China · Jun.-Aug. 2025

Research Intern · Advisor: Jiwen Zhang (Associate Research Fellow)

Quasi-Real-Time Digital Twin for Robotic Assembly

- Built a ROS 2 non-blocking pipeline separating real-time safety control from background FEA.
- Used kinematic TIE constraints and mesh-scale analysis to improve robustness and computational predictability.
- Integrated full-field stress outputs into an FSM-based safety evaluator.

SKILLS

Programming
Python, Java, MATLAB, ROS 2

Robotics & Control
Robot modeling and control, grasp planning, safety reflex control, FSM decision logic

Learning & Perception
Reinforcement learning, YOLOv5, MediaPipe, random forests

Simulation & Modeling
MuJoCo/MJLab, FEA/FEM, rigid-soft contact, mesh-scale analysis

Design & Fabrication
CAD assemblies, PneuNet grippers, high-density FES sleeves

HONORS & AWARDS

First Prize (Provincial Level, 2nd Place), Guangdong Provincial College Students Engineering Practice and Innovation Competition — 2025

Outstanding Student Scholarship — 2024-2025

Outstanding Student Scholarship — 2023-2024